# Unit 3.2 The product and quotient rules

Note Title 2015/03/22

Product rule used for eg $f(x) = x^2 \cdot e^x$

Quotient rule used for eg $f(x) = \frac{x^2}{e^x}$ $\qquad \frac{e^x \cdot \sin x}{x^2}$

## HOW IT WORKS

### Product rule:

$$\frac{d}{dx}\left(f(x) \cdot g(x)\right) = f'(x)\, g(x) + f(x)\, g'(x)$$

eg. $\frac{d}{dx}\left(x^2 e^x\right) = 2xe^x + x^2 e^x$

$$\frac{d}{dx}\left(e^x\left(\sqrt{x}+1\right)\right) = e^x\left(\sqrt{x}+1\right) + e^x \cdot \frac{1}{2}x^{-1/2}.$$

$$\frac{d}{dx}\left(5x^5 \cdot \left(x^4 + x\right)\right) = 25x^4\left(x^4+x\right) + 5x^5\left(4x^3+1\right).$$

### Quotient rule

$$\frac{d}{dx}\left(\frac{f(x)}{g(x)}\right) = \frac{f'(x)\,g(x) - f(x)\,g'(x)}{\left(g(x)\right)^2}$$

eg. $\frac{d}{dx}\left(\frac{x^4}{x^2+2x}\right) = \frac{4x^3\left(x^2+2x\right) - x^4\left(2x+2\right)}{\left(x^2+2x\right)^2}$

$$\frac{d}{dx}\left(\frac{4x^3 - 2x^2 + \frac{1}{x}}{e^x}\right)$$

$$= \frac{\left(12x^2 - 4x - \frac{1}{x^2}\right)e^x - \left(4x^3 - 2x^2 + \frac{1}{x}\right).e^x}{(e^x)^2}$$

Special quotient rule:

$$\frac{d}{dx}\left(\frac{1}{g(x)}\right) = \frac{-g'(x)}{(g(x))^2}$$

eg $$\frac{d}{dx}\left(\frac{1}{e^x + x}\right) = \frac{-(e^x+1)}{(e^x+x)^2}$$

$$\frac{d}{dx}\left(\frac{1}{4 - x^{10}}\right) = \frac{10x^9}{(4-x^{10})^2}$$

Why the product rule works:

$$\frac{d}{dx}(f(x).g(x))$$

$$= \lim_{h \to 0} \frac{f(x+h)g(x+h) - f(x).g(x)}{h}$$

$$= \lim_{h \to 0} \frac{f(x+h)g(x+h) - f(x)g(x+h) + f(x)g(x+h) - f(x).g(x)}{h}$$

$$= \lim_{h \to 0} \frac{f(x+h) - f(x)}{h} \times g(x+h) + \lim_{h \to 0} f(x)\left[\frac{g(x+h) - g(x)}{h}\right]$$

$$= f'(x)\,g(x) + f(x) \cdot g'(x).$$

Mixed examples

$$\frac{d}{dx}\left(\frac{x^2 \cdot e^x}{\sqrt{x}+1}\right) \begin{matrix} f \\ g \end{matrix} = \frac{(2x \cdot e^x + x^2 e^x)(\sqrt{x}+1) - x^2 \cdot e^x \cdot \frac{1}{2} x^{-\frac{1}{2}}}{(\sqrt{x}+1)^2}$$

$$\frac{d}{dx} \frac{4x^2 - x}{x^3 e^x} = \frac{(8x-1)x^3 e^x - (4x^2 - x)(3x^2 e^x + x^3 e^x)}{(x^3 e^x)^2}$$

$$\frac{d}{dx}\left(x^2 \cdot e^x (\sqrt{x}+1)\right)$$

$$= 2x \cdot e^x (\sqrt{x}+1) + x^2 \cdot e^x (\sqrt{x}+1) + x^2 \cdot e^x \left(\frac{1}{2} x^{-1/2}\right).$$

$$\frac{d}{dx}\left(\frac{2x^3 \cdot e^x + 4}{x^2 \cdot e^x}\right)$$

$$= \frac{(6x^2 \cdot e^x + 2x^3 \cdot e^x)(x^2 \cdot e^x) - (2x^3 \cdot e^x + 4)(2xe^x + x^2 e^x)}{(x^2 \cdot e^x)^2}$$